



M3RAG: Orchestrating Multi-agent Reasoning for Multi-hop, Multi-modal Understanding

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Abstract. Traditional Retrieval-Augmented Generation (RAG) systems face significant challenges in processing visually-rich, multi-modal documents, especially when addressing complex queries that demand deep semantic understanding and cross-document reasoning. To bridge this gap, we propose a novel framework M3RAG (Multi-hop, Multi-modal, Multi-agent Retrieval-Augmented Generation). M3RAG introduces a collaborative multi-agent architecture that orchestrates an iterative process for query planning, fine-grained multi-modal evidence extraction, and self-correcting answer verification. By dynamically planning information retrieval pathways and fusing cross-modal evidence, M3RAG enables robust reasoning for complex multi-hop questions and achieves a deeper understanding of multi-modal content. Extensive experimental results demonstrate that M3RAG establishes a new state-of-the-art, outperforming existing methods by up to 18.4% on challenging multi-hop, multi-modal document understanding benchmarks, validating its effectiveness in complex reasoning tasks.

1 Introduction

Large Language Models (LLMs) [1,2] demonstrated remarkable capabilities in various natural language processing tasks [3,4]. However, they often struggle with factual inaccuracies and the assimilation of up-to-date knowledge [5,6]. Retrieval-Augmented Generation (RAG) [7–10] addresses these limitations by enabling LLMs to access and integrate external knowledge bases, thereby mitigating hallucinations and expanding their knowledge coverage.

Despite the advancements in RAG, current systems predominantly operate on text-based contexts, which inherently limit their application in real-world scenarios where information is frequently embedded within visually-rich documents [11,12]. These documents, such as charts, tables, and images interleaved with text, pose significant challenges for traditional RAG pipelines. Relying on optical character recognition (OCR) and parsing processes to extract text can lead to crucial information loss, including visual cues and layout, and introduce errors that propagate through retrieval and generation stages [13]. While the emergence of Vision-Language Models (VLMs) offers a promising avenue

to overcome these limitations, extending RAG to vision-language tasks presents unique challenges such as modality discrepancy and information noise.

While existing works have begun to address multi-modal RAG [9, 13–16], these approaches primarily aim to overcome the limitations of traditional RAG in processing visually-rich documents. For instance, frameworks like VisRAG and VDocRAG have demonstrated the benefits of encoding and retrieving documents directly as images to maximize information retention and avoid parsing losses. However, a critical research gap persists as these frameworks have yet to effectively scale to complex, multi-hop queries. Answering such queries in real-world scenarios necessitates gathering and synthesizing information from multiple distinct documents or different sections within a document, thereby forming a multi-hop logical chain. Filling this gap is particularly difficult due to two key challenges. First, the inherent complexity of multi-hop reasoning itself requires a dynamic and structured approach to decompose complex queries and retrieve disparate pieces of evidence. Second, this complexity is significantly amplified by modality discrepancy and retrieval noise, which are often magnified in a multi-hop setting and can compromise the entire reasoning chain.

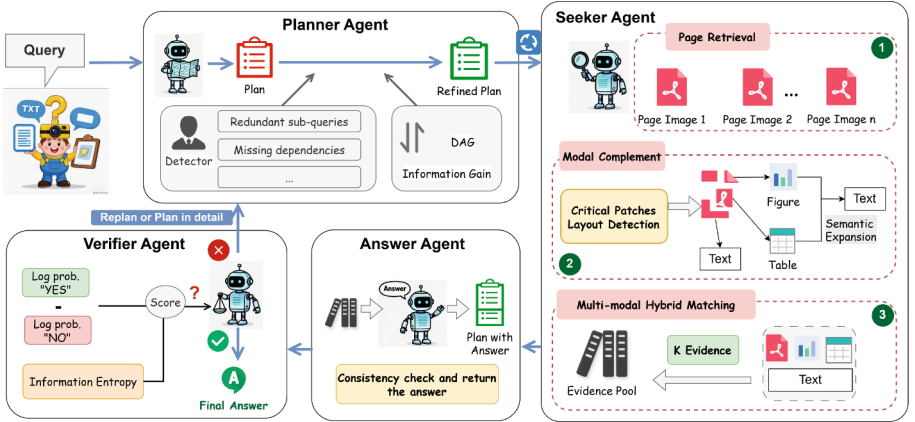


Fig. 1. The overview of M3RAG framework.

To address this critical gap and to effectively tackle the challenges of multi-hop reasoning within complex visual documents, this paper proposes a novel framework, **M3RAG** (**Multi-hop, Multi-modal, Multi-agent** Retrieval-Augmented Generation). Building upon advancements in vision-based RAG, M3RAG uniquely integrates a multi-agent architecture that orchestrates a sophisticated, iterative process for comprehensive query understanding and precise evidence synthesis. This framework is designed to: (1) enable robust multi-hop reasoning over visually-rich documents by dynamically planning information retrieval pathways and decomposing complex queries into manageable sub-goals; (2) achieve a deeper understanding of multi-modal content through tar-

geted regional analysis and cross-modal evidence fusion, moving beyond simple page-level retrieval; and (3) enhance overall answer accuracy and reliability via a self-correcting, iterative verification mechanism that provides a crucial feedback loop. Our approach represents a significant step towards empowering LLMs to accurately and robustly answer complex, real-world questions from diverse, multi-modal knowledge sources.

Our **main contributions** are summarized as follows:

- We propose M3RAG, a novel multi-hop, multi-modal, multi-agent RAG framework by integrating specialized agents to orchestrate complex reasoning over visually-rich documents.
- We introduce fine-grained multi-modal evidence extraction through novel techniques like regional re-reading and modality expansion, which significantly reduce noise and enhance the precision of information retrieval from visually-rich documents.
- We develop a dynamic multi-agent collaborative and iterative reasoning mechanism for multi-hop problem-solving, which includes intelligent query planning and a self-correcting verification feedback loop, ensuring robust and reliable answers.
- We demonstrate superior performance through extensive experiments on challenging multi-hop, multi-modal question answering benchmarks, where M3RAG significantly outperforms existing state-of-the-art RAG methods.

2 Related Work

2.1 Multi-Modal RAG and Visual Document Understanding

Traditional RAG systems, primarily text-based [11], struggle with visually-rich documents due to information loss from OCR and neglected visual cues. To overcome this, multi-modal RAG paradigms have emerged. Approaches such as VisRAG [13], ViDoRAG [15], and VDocRAG [9] directly encode documents as images, leveraging Vision-Language Models for comprehensive visual information retention in retrieval and generation. M3DoCRAG [16] demonstrated a multi-modal RAG framework for multi-page, multi-document understanding. However, despite these advancements [14, 17] in integrating visual information and processing multi-page documents, prior multi-modal RAG systems have not specifically introduced mechanisms for orchestrated multi-hop reasoning or fine-grained, iterative evidence refinement across distributed multi-modal sources. Their primary focus has been on comprehensive visual encoding and retrieval, rather than the intricate flow control needed for complex multi-hop inference.

2.2 Multi-Agent Frameworks for Question Answering and RAG

The increasing complexity of real-world QA tasks necessitates multi-agent frameworks in RAG systems [18, 19]. Various multi-agent RAG approaches enhance

reasoning and mitigate limitations. MA-RAG [20] focuses on collaborative reasoning for complex retrieval and multi-hop QA. M-RAG [21] optimizes system-level performance and memory refinement via multi-partitioning and reinforcement learning. MAIN-RAG [7] significantly improves noise robustness using an unrelated multi-agent filtering mechanism. Beyond core RAG functions, specialized frameworks like ATM [22] offer adversarial tuning for manipulation resistance. Despite these advancements, most existing multi-agent RAGs primarily address textual information or feature simplistic visual integration. As discussed in Sect. 2.1, many current multi-modal RAG systems still lack dedicated mechanisms for fine-grained, iterative evidence refinement across diverse modalities, or the intricate flow control necessary for complex multi-hop inference. The M3RAG uniquely bridges this gap by combining a heterogeneous multi-agent architecture with fine-grained multi-modal evidence retrieval and a dynamic self-verification loop.

3 Method

This section presents a comprehensive overview of our proposed M3RAG framework, designed to tackle the complexities of multi-hop question answering over visually-rich, multi-document corpora. As depicted in Fig. 1, M3RAG leverages a collaborative multi-agent architecture to orchestrate a dynamic, iterative reasoning process, integrating multi-modal information extraction and a self-correcting verification mechanism.

3.1 Definition of LLM Agents in M3RAG

The M3RAG framework employs a robust multi-agent architecture with a heterogeneous design, where each agent is powered by a different type or size of model, ranging from LLMs and VLMs to smaller, more specialized models. Each agent is designed with a distinct role to facilitate complex multi-hop question answering over visually-rich, multi-document corpora. These agents collaborate in an intelligent iterative loop, as shown in Algorithm 1, to ensure systematic reasoning, precise evidence acquisition, and reliable answer generation.

Planner ($\mathcal{A}_P$). The **Planner** agent is the strategic orchestrator of the multi-hop reasoning process. It logically decomposes a complex user query Q into a series of smaller, interdependent sub-queries. This process can be described as a function that maps the input query Q to an optimized execution plan P : $\mathcal{A}_P : Q \rightarrow P$.

Seeker ($\mathcal{A}_S$). The **Seeker** agent is dedicated to the precise and robust retrieval of multi-modal evidence from the document corpus $\mathcal{D}$. Guided by a sub-query q_j from the Planner’s plan, $\mathcal{A}_S$ identifies and extracts highly relevant textual or visual information. The Seeker’s role can be formally represented as a function $\mathcal{A}_S$ that takes a sub-query q_j and the corpus $\mathcal{D}$ as input to produce a set of relevant evidence $\mathcal{E}_j$: $\mathcal{A}_S(q_j, \mathcal{D}) \rightarrow \mathcal{E}_j$.

Answer ($\mathcal{A}_A$). The **Answer** agent is responsible for synthesizing a coherent, concise, and accurate response to the original user query Q . Its objective is to leverage all gathered evidence from the Seeker to formulate a definitive answer A . The Answer agent's operation can be viewed as a function that combines the original query Q and the evidence pool $\mathcal{E}$ to generate a final answer A : $\mathcal{A}_A : (Q, \mathcal{E}) \rightarrow A$.

Verifier ($\mathcal{A}_V$). The **Verifier** agent acts as the critical quality control and feedback mechanism. Its central role is to rigorously evaluate the correctness, completeness, and factual consistency of the answer generated by $\mathcal{A}_A$. This function takes the draft answer A , original query Q , and evidence pool $\mathcal{E}$ as input, and returns a confidence score C and a potential feedback signal F : $\mathcal{A}_V : (A, Q, \mathcal{E}_{pool}) \rightarrow (C, F)$.

Algorithm 1. M3RAG Collaborative Algorithm

Require: User Query Q , Multi-modal Document Corpus $\mathcal{D}$, Maximum Iterations $MaxIter$

Ensure: Final Answer A^*

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1: Initialize:  $P \leftarrow \text{None}$ ,  $iteration \leftarrow 0$ ,  $A^* \leftarrow \text{None}$ ,  $F \leftarrow \text{REPLAN}$     //Initialize
   feedback signal to trigger initial planning
2: while  $iteration < MaxIter$  and  $A^*$  is None do
3:   if  $F$  is REPLAN then
4:      $P_{plan} \leftarrow \mathcal{A}_P(Q)$ 
5:   else if  $F$  is REFINES then
6:      $P_{plan} \leftarrow \mathcal{A}_P(\text{Refine}, Q, \text{Signal} = F)$ 
7:   end if    //For CONTINUE signal,  $P_{plan}$  remains unchanged, no Planner call.
8:    $\mathcal{E} \leftarrow \emptyset$ 
9:   for each sub-query  $q_j$  in  $P_{plan}$  do
10:     $\mathcal{E}_j \leftarrow \mathcal{A}_S(q_j, \mathcal{D})$ 
11:     $\mathcal{E} \leftarrow \mathcal{E} \cup \mathcal{E}_j$ 
12:   end for
13:    $A \leftarrow \mathcal{A}_A(Q, \mathcal{E})$     //Generate a draft answer
14:    $C, F_{next} \leftarrow \mathcal{A}_V(A, Q, \mathcal{E})$     //Verify answer and get next feedback signal
15:   if  $F_{next}$  is ACCEPT then
16:      $A^* \leftarrow A$     //Answer is satisfactory, set as final answer
17:   end if
18:    $F \leftarrow F_{next}$     //Update feedback signal for the next iteration
19:    $iteration \leftarrow iteration + 1$ 
20: end while
21: Return  $A^*$ 

```

3.2 Fine-Grained Multi-modal Evidence Extraction

Regional Re-reading. Initial page-level retrieval often introduces noise, as much of a page may be irrelevant to a specific sub-query. The **Regional Re-reading** mechanism mitigates this by focusing on semantically salient regions within retrieved pages.

1. **Page Partitioning:** For a retrieved page p_i , we leverage the underlying VLM’s vision encoder (*e.g.*, MiniCPM-V 2.6’s) to partition it into a set of visual patches (tokens) $\mathcal{P}_i = \{r_{i,1}, \dots, r_{i,S}\}$. Each patch $r_{i,s}$ represents a localized area of the page.
2. **Relevance Scoring:** To determine the relevance of each patch, we leverage a VLM’s shared embedding space. The relevance score for a patch $r_{i,s}$ with respect to the sub-query q_j is calculated as the cosine similarity between their respective embeddings:

$$\text{score}(r_{i,s}, q_j) = \text{Sim}(\text{Embed}(r_{i,s}), \text{Embed}(q_j)), \quad (1)$$

where $\text{Embed}(\cdot)$ maps visual regions and text queries into a shared embedding space.

3. **Patch Selection:** From the ranked list of patches, we select the top- m most relevant patches. This refined set of regions, $\mathcal{R}_j \subseteq \mathcal{P}_i$, effectively reduces visual noise and focuses on key areas for subsequent analysis.

Modality Expansion. Crucial information in visually-rich documents is often embedded in non-textual elements. **Modality Expansion** extracts these visual insights and converts them into a textual format, thereby enriching the evidence pool.

1. **Visual Element Identification:** Within the selected relevant regions $r_s \in \mathcal{R}_j$, we identify specific visual elements $V_{elem} = \{v_1, \dots, v_L\}$, such as figures and tables.
2. **Textual Description Generation:** For each visual element $v_l \in V_{elem}$, a VLM, denoted as VLM_{desc} , generates a detailed textual description d_l . The generation process is:

$$d_l = \text{VLM}_{\text{desc}}(v_l, q_j, \text{Context}(v_l)), \quad (2)$$

where $\text{Context}(v_l)$ refers to relevant surrounding text. This transforms complex visual information into a semantic text representation.

Evidence Consolidation and Ranking. The generated textual descriptions $\{d_l\}_{l=1}^L$ are combined with the original textual content from the selected regions. All types of evidence fragments are then embedded into a shared space [23], ranked by relevance to q_j , and the top- K pieces are selected to form the final evidence pool $\mathcal{E}$.

3.3 Plan Generation and Self-Verification for Multi-Hop Processing

Plan Generation for Multi-Hop Queries. The Planner agent ($\mathcal{A}_P$) initiates the multi-hop reasoning process by systematically decomposing the user’s high-level multi-hop query Q into a sequence of actionable and interdependent sub-queries.

Formally, given an input query Q , $\mathcal{A}_P$ first generates an initial execution plan P by thinking step-by-step. This plan is a Directed Acyclic Graph (DAG) $G = (V, E)$. Each node $v_j \in V$ corresponds to a distinct sub-query q_j , and directed edges $(v_i, v_j) \in E$ represent dependencies where q_j relies on the successful resolution of q_i . The plan P can be initially represented as:

$$P = \mathcal{A}_P(Q) = \{(q_j, \text{Dependencies}(q_j))\}_{j=1}^N, \quad (3)$$

where $\text{Dependencies}(q_j)$ denotes the set of sub-queries that must be resolved prior to q_j .

Following the initial generation, a self-judgment **Detector** inspired by the principles of Agent-Oriented Programming [24] is invoked to ensure the plan’s reliability. The plan is accepted only if it satisfies the **Consistency**, **Non-redundancy**, and **Dependency Completeness**.

To further optimize the plan, the sub-queries are prioritized based on their expected information gain. We quantify this using an information entropy measure $H(q_j)$, where a sub-query with higher entropy is considered to contain more uncertainty and thus represents a more critical piece of information to acquire. The final execution order is determined by a topological sort, which satisfies all dependencies, and then a prioritization of sub-queries with the highest entropy values. The entropy of a sub-query q_j is defined as:

$$H(q_j) = - \sum_{w \in \mathcal{V}_j} p(w|q_j) \log p(w|q_j), \quad (4)$$

where $\mathcal{V}_j$ is the vocabulary of the sub-query q_j , and $p(w|q_j)$ is the probability of a term w appearing given the sub-query context.

Self-Verification and Iterative Refinement. The Verifier agent ($\mathcal{A}_V$) serves as a crucial internal feedback loop for $\mathcal{A}_P$, providing a robust mechanism for assessing answer quality and guiding subsequent refinement. This scheme, inspired by reward models and adaptive filtering RAG frameworks [7], fuses a confidence score with an entropy-based change metric to make dynamic, data-driven decisions, thereby reducing the reliance on multiple static thresholds.

Formally, at each iteration t , $\mathcal{A}_V$ computes a confidence score C_t for a generated answer A_t against retrieved evidence $\mathcal{E}_t$. This score quantifies the LLM’s probabilistic belief in the answer’s correctness:

$$C_t = P_{\text{yes}}(\text{Correct}|A_t, \mathcal{E}_t) - P_{\text{no}}(\text{Correct}|A_t, \mathcal{E}_t). \quad (5)$$

The decision logic is not predicated on a fixed confidence range but on the dynamic behavior of this score, specifically its change rate, $\Delta C_t = |C_t - C_{t-1}|$,

which acts as a proxy for the knowledge entropy gained in the current iteration. The process is governed by a single final confidence threshold τ_{final} . If the confidence score C_t reaches this threshold, the answer is deemed reliable and the process terminates. Otherwise, if $C_t < \tau$ but the change rate ΔC_t is high, it signifies that the current plan is still yielding valuable information, and the system proceeds with the next sub-task. Conversely, if $C_t < \tau$ and ΔC_t remains negligible for a predefined number of consecutive iterations, it indicates stagnation, and the Verifier requests the Planner to generate a new or revised plan to explore alternative reasoning paths. The entire process is also subject to a maximum number of iterations, at which point it is forcibly terminated.

4 Experiments

4.1 Experimental Setup

Datasets. We evaluate M3RAG on a comprehensive suite of seven challenging multi-modal Visual Question Answering datasets. **ArxivQA**, which focuses on scientific papers demanding reasoning across figures, tables, and dense text; **ChartQA** and **PlotQA**, comprising questions over various types of charts and plots, necessitating visual understanding and data extraction; **DocVQA**, a broad dataset covering general document understanding, including forms, invoices, and reports; **InfoVQA**, featuring documents with a strong emphasis on infographics and structured information extraction; and **SlideVQA**, containing questions derived from presentation slides, testing comprehension of structured and visual layouts. Additionally, we include **ViDoSeek** [15], a recent benchmark specifically designed to evaluate RAG performance on visually rich documents requiring complex, often multi-hop, reasoning across visual elements. This diverse set ensures a thorough assessment of M3RAG’s capabilities in real-world scenarios.

Evaluation Metrics. For our end-to-end evaluation, we employed a model-based assessment using GPT-4o, which involved assigning scores from 1 to 10 by comparing the reference answer with the final answer. Answers receiving scores of 7 or above were considered correct, and we subsequently calculated accuracy as the evaluation metric.

Baselines. We compare M3RAG against several RAG approaches for visually-rich documents and Text-RAG [15]. **VisRAG** [13] pioneers a vision-based RAG pipeline, directly embedding documents as images for retrieval and generation. **ViDoRAG** [15] is a multi-agent RAG framework for complex reasoning in visual documents. **VDocRAG** [9] integrates visual document understanding into RAG, enhancing retrieval and generation via joint text-visual embeddings. The **Oracle** serves as the upper bound performance, where the model responds based on the golden page.

Implementation Details. For our experiments, we utilize two powerful Vision-Language Models as our base VLMs, serving as the backbone for Seeker

and Answer within M3RAG: **MiniCPM-V 2.6** [25] and **Qwen2.5-VL-7B-Instruct** [26]. The Planner and Verifier employ the LLM **Mistral-7B** [2] as backbone. All experiments are run 5 times (we report mean accuracy) on 2 NVIDIA A100 GPUs. During retrieval, we select Top-5 candidate pages [13]. For each page, we select $m=144$ visual tokens that are the most related to the query [27].

4.2 Retrieval-Augmented Generation Results

Table 1. Comprehensive evaluation of M3RAG against various baselines on multi-modal VQA datasets. Results are reported in Accuracy (%). **Oracle** represents the upper bound of performance. **Text-RAG** denotes a traditional text-based RAG approach. **Bold** entries highlight the current SOTA performance among baselines.

Methods	ArxivQA	ChartQA	DocVQA	InfoVQA	PlotQA	SlideVQA	ViDoSeek
<i>Base VLM: MiniCPM-V 2.6</i>							
Direct	29.10 (40.9)	31.05 (38.9)	40.58 (48.7)	35.45 (48.1)	19.86 (31.7)	24.68 (36.4)	17.69 (24.3)
Text-RAG	47.12 (66.30)	42.38 (53.07)	58.15 (69.85)	50.43 (68.47)	32.91 (52.50)	37.88 (55.93)	40.21 (55.25)
VDocRAG	67.92 (95.55)	68.55 (85.85)	74.88 (89.95)	59.90 (81.33)	49.70 (79.28)	56.95 (84.08)	61.82 (84.94)
VisRAG	67.77 (95.3)	53.97 (67.6)	70.90 (85.2)	54.46 (73.9)	38.93 (62.1)	50.72 (74.9)	49.67 (68.2)
ViDoRAG	68.80 (96.8)	69.70 (87.3)	75.65 (90.9)	60.55 (82.2)	50.64 (80.8)	57.73 (85.2)	62.58 (86.0)
Oracle	71.08 (100.0)	79.85 (100.0)	83.25 (100.0)	73.65 (100.0)	62.69 (100.0)	67.73 (100.0)	72.78 (100.0)
M3RAG	69.23 (97.4)	80.95 (101.4)	80.91 (97.2)	67.47 (91.6)	59.49 (94.9)	64.32 (94.9)	67.82 (93.2)
	(↑10.6%)	(↑16.1%)	(↑6.9%)	(↑11.4%)	(↑17.5%)	(↑11.4%)	(↑8.4%)
<i>Base VLM: Qwen2.5-VL-7B-Instruct</i>							
Direct	30.37 (41.0)	34.30 (42.4)	44.40 (52.0)	37.87 (53.9)	21.23 (30.8)	24.78 (35.4)	19.34 (24.8)
Text-RAG	51.35 (69.30)	48.72 (60.23)	61.58 (72.19)	53.29 (75.87)	38.05 (55.22)	41.22 (58.91)	43.19 (55.44)
VDocRAG	69.15 (93.32)	69.51 (85.93)	70.87 (83.06)	62.82 (89.41)	56.88 (82.58)	58.73 (83.89)	61.47 (78.95)
VisRAG	62.97 (85.0)	56.66 (70.1)	68.98 (80.9)	54.55 (77.7)	48.05 (69.7)	52.28 (74.7)	49.67 (63.8)
ViDoRAG	69.98 (94.4)	70.20 (86.8)	71.75 (84.1)	63.45 (90.3)	57.49 (83.4)	59.38 (84.9)	62.18 (79.8)
Oracle	74.10 (100.0)	80.88 (100.0)	85.30 (100.0)	70.24 (100.0)	68.90 (100.0)	69.97 (100.0)	77.90 (100.0)
M3RAG	70.26 (94.8)	78.75 (97.4)	80.06 (93.9)	68.67 (97.8)	64.66 (93.8)	65.82 (94.1)	69.36 (89.0)
	(↑0.4%)	(↑12.2%)	(↑11.6%)	(↑8.2%)	(↑12.5%)	(↑10.9%)	(↑11.6%)

As shown in Table 1, M3RAG framework consistently achieves state-of-the-art (SOTA) performance across all tested multi-modal VQA datasets, surpassing existing single-agent RAG methods by a significant margin. This success highlights the effectiveness of our multi-agent architecture in handling the complexities of visually-rich, multi-hop question answering. The most substantial gains are observed on datasets requiring fine-grained visual reasoning, such as ChartQA, PlotQA, and ViDoSeek. This is because these datasets heavily rely on fine-grained visual information extraction and numerical reasoning, which are challenging for traditional text-based RAG and even some single-agent multi-modal approaches. The multi-agent design of M3RAG, with its specialized Seeker

for multi-modal evidence retrieval and Verifier for iterative refinement, is better equipped to handle these intricate visual details and structured data. While other multi-modal baselines like ViDoRAG also show strong performance, especially on datasets with simpler document structures, they are less effective when faced with complex data integration or multi-hop reasoning. This suggests that their single-agent pipeline, while a step up from text-only methods, lacks the dynamic and collaborative problem-solving capabilities of a multi-agent system.

4.3 Analysis

Ablations. Table 2 presents the results of our ablation study, demonstrating the critical role each component plays in M3RAG’s superior performance. Removing the Planner leads to a noticeable decline in accuracy, especially on datasets requiring complex multi-hop reasoning, underscoring its importance in structuring the problem-solving process. Similarly, ablating Modal Expansion results in a significant performance drop, particularly on datasets rich in visual information, highlighting its necessity for effectively leveraging non-textual content. The absence of Regional Re-reading causes the most substantial performance degradation across all datasets, confirming its foundational role in precisely extracting relevant information and filtering out noise from dense multi-modal pages. Lastly, removing the Verifier also negatively impacts accuracy, emphasizing its crucial function in iterative refinement and self-correction, which is vital for achieving the highest levels of robustness and precision. These results collectively validate that each component of M3RAG contributes significantly to its overall effectiveness in complex multi-modal document understanding.

Table 2. Ablation study of our method on MiniCPM-V 2.6 (Accuracy %)

Methods	ArxivQA	ChartQA	PlotQA	SlideVQA	ViDoSeek
M3RAG (Full)	69.23	80.95	59.49	64.32	67.82
w/o Planner	66.71	77.40	56.83	60.15	64.90
w/o Modal Expansion	67.05	78.12	55.97	59.28	62.47
w/o Regional Re-reading	65.34	76.58	54.26	57.41	61.03
w/o Verifier	67.89	79.03	57.62	61.75	65.34

How Do Different Evidence Types Contribute to Generation? Figure 2 illustrates the proportional contribution of various evidence types under correct and incorrect prediction scenarios. For successful answers, M3RAG primarily leverages Regional Textual (including Modality Expanded) and Regional Visual/Structured evidence, indicating its proficiency in extracting and utilizing fine-grained information. Conversely, in cases where retrieval is accurate but the answer is not, the contribution from Whole Page Context significantly increases, suggesting that an over-reliance on coarse-grained, noisy information can hinder precise reasoning. This highlights the critical role of M3RAG’s refined extraction and filtering mechanisms for robust performance.

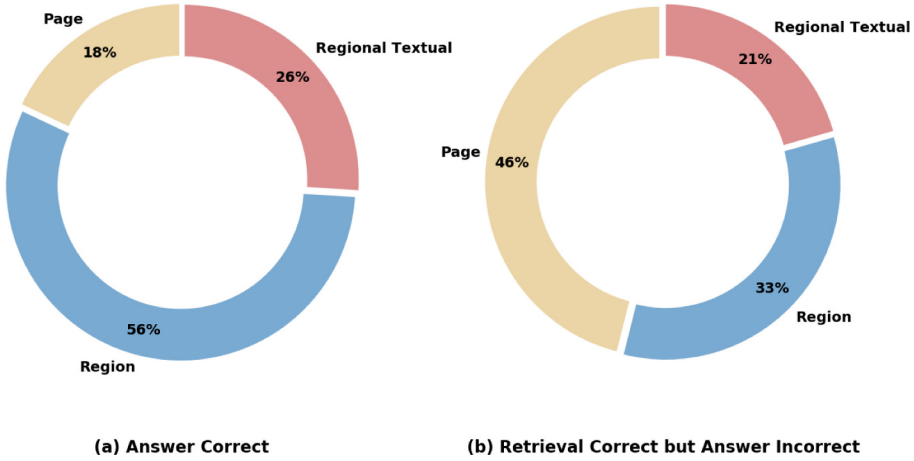


Fig. 2. Root causes of correct and incorrect predictions.

Which Types of Problems is Our Method More Suitable for Handling?

Figure 3 provides a comprehensive overview of M3RAG’s performance across multiple critical dimensions. M3RAG consistently demonstrates superior performance, covering a larger area on the radar chart, which signifies its strong overall capability. This qualitative advantage is particularly pronounced in crucial aspects like Multi-hop Reasoning, Multi-modal Integration, and Accuracy. The collaborative nature of our multi-agent framework allows for the systematic breakdown of complex queries, resulting in a significantly stronger performance on multi-hop reasoning tasks. Furthermore, the cross-modal hybrid matching mechanism proves instrumental in effectively fusing information from disparate sources, directly contributing to our pronounced advantage in multi-modal integration. This confirms the effectiveness of our iterative planning and cross-modal hybrid matching, enabling M3RAG to robustly tackle complex multi-hop and multi-modal information fusion tasks.

How Does Our Method Perform in Time Efficiency?

Figure 4 presents a detailed analysis of the average inference latency for M3RAG and baseline methods, broken down by component. As observed, M3RAG generally requires more time compared to basic, non-agent frameworks like Direct and VisRAG. Within our multi-agent framework, the Seeker agent, responsible for extensive multi-modal retrieval, regional re-reading, and modality expansion, consistently accounts for the largest portion of the total inference time. Notably, M3RAG exhibits lower overall latency compared to other multi-agent frameworks such as ViDoRAG. Furthermore, a crucial observation is the efficiency gain achieved by the Verifier agent: when the Verifier is integrated, M3RAG’s overall latency is reduced compared to the ‘M3RAG w/o Verifier’ variant. This counterintuitive reduction in time, despite the Verifier’s own execution, is attributed to its ability

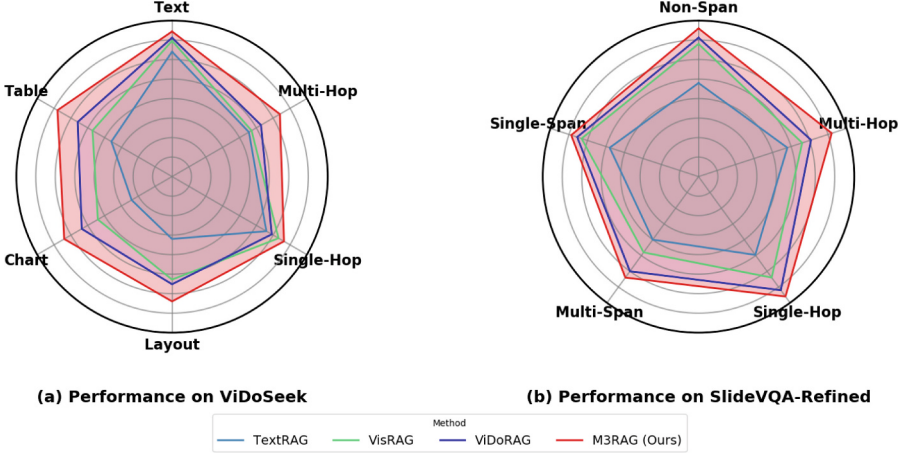


Fig. 3. Performance across different types of queries on ViDoSeek and the refined SlideVQA datasets. The outer boundary of the radar chart represents the Oracle’s performance, indicating the theoretical upper bound.

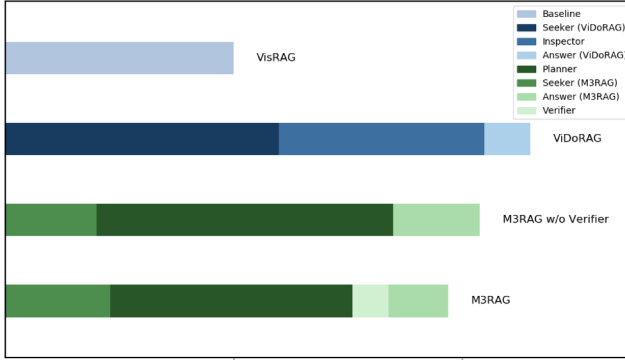


Fig. 4. Latency analysis and component breakdown of M3RAG and baselines. VisRAG serves as a reference for total time consumption.

to significantly decrease the number of unnecessary iterations, thereby leading to faster convergence to a correct answer and ultimately improving both accuracy and time efficiency.

How Does Our Method Perform Under Different Hyperparameters?

Our sensitivity analysis in Fig. 5 demonstrates the balanced behavior of M3RAG under varying hyperparameter settings. For the confidence threshold τ , we found that while a low value fails to effectively verify answers, an excessively high τ causes the Verifier to repeatedly reject responses, increasing iteration counts. The necessity of a maximum iteration limit (*i.e.*, 7) to prevent infinite loops likely

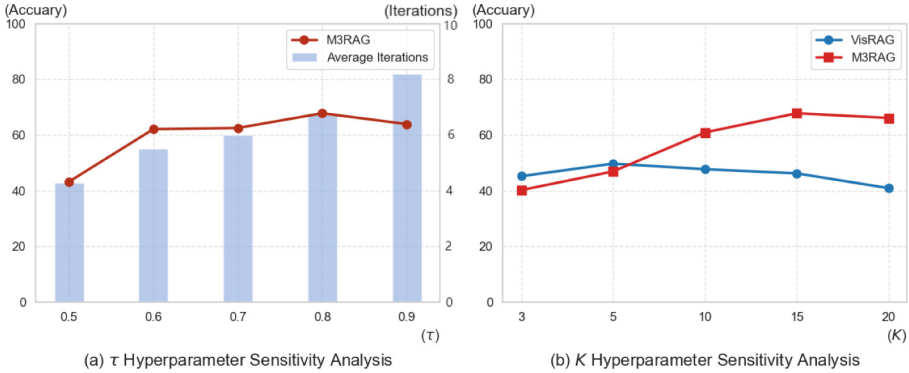


Fig. 5. Performance under different hyperparameters on ViDoSeek.

contributes to the slight performance drop observed at the highest threshold. Regarding the number of candidate evidence patches K , our system achieves peak performance with an optimal number of patches; too few can lead to missing crucial information, while too many introduce noise. Compared to VisRAG, M3RAG’s use of more fine-grained evidence necessitates and effectively utilizes a larger optimal K value.

5 Conclusion

In this paper, we present M3RAG, a novel multi-agent RAG framework for complex multi-hop question answering on visually-rich documents. Our work’s core contributions include a heterogeneous multi-agent architecture that orchestrates reasoning and a dynamic iterative refinement mechanism that enables self-correction. We also introduced fine-grained multi-modal retrieval techniques to effectively process intricate visual and textual evidence. Our experiments show that M3RAG consistently outperforms existing single-agent RAG baselines, especially on datasets requiring deep visual understanding and multi-hop reasoning. The results confirm that an integrated multi-agent system provides a more robust and accurate approach to complex document understanding.

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